



Dynamic Enrichment for Evaluation of Protein Networks (DEEPN): A High Throughput Yeast Two-Hybrid (Y2H) Protocol to Evaluate Networks

Ali Zeeshan Fakhar, Jinbao Liu, and Karolina M. Pajerowska-Mukhtar

Abstract

Proteins are the building blocks of life, and a vast array of cellular processes is handled by protein–protein interactions (PPIs). The protein complexes formed via PPIs lead to tangled networks that, with their continuous remodeling, build up systematic functional units. Over the years, PPIs have become an area of interest for many researchers, leading to the development of multiple *in vitro* and *in vivo* methods to reveal these interactions. The yeast-two-hybrid (Y2H) system is a potent genetic way to map PPIs in both a micro- and high-throughput manner. Y2H is a technique that involves using modified yeast cells to identify protein–protein interactions. For Y2H, the yeast cells are engineered only to grow when there is a significant interaction between a specific protein with its interacting partner. PPIs are identified in the Y2H system by stimulating reporter genes in response to a restored transcription factor. However, Y2H results may be constrained by stringency requirements, as the limited number of colony screenings through this technique could result in the possible elimination of numerous genuine interactions. Therefore, DEEPN (dynamic enrichment for evaluation of protein networks) can be used, offering the potential to study the multiple static and transient protein interactions in a single Y2H experiment. DEEPN utilizes next-generation DNA sequencing (NGS) data in a high-throughput manner and subsequently applies computational analysis and statistical modeling to identify interacting partners. This protocol describes customized reagents and protocols through which DEEPN analysis can be utilized efficiently and cost-effectively.

Key words DEEPN, Y2H, PPI, Illumina Sequencing, System Biology, High-throughput

1 Introduction

Proteins, sometimes called “molecular machines,” function by intricate dynamic interactors with other complex molecules such as DNA, RNA, and other proteins in living systems [1–9]. Multiple processes in living systems, including signaling, molecular trafficking, immunomodulation, metabolic activities, and further growth and developmental regulations, are accurately governed via proteins or protein–protein interactions (PPIs) [10]. Many *in vitro*

and *in vivo* techniques have been developed over time for studying PPIs. Biotechnological advancements make these newly devised methods readily available to biologists/researchers. Information into the system for potential new binding partners typically precedes a study of protein–protein interactions. Affinity-tagged proteins/affinity-based protein profiling, Y2H, and other quantitative proteomic-based approaches that can be combined, such as co-immunoprecipitation and affinity chromatography, can be utilized to study the potential PPIs [11–19]. Several techniques can validate the PPIs, such as co-immunoprecipitation, surface plasmon resonance (SPR), confocal microscopy for intracellular protein colocalization, and spectroscopic studies [20]. Despite being widely used, several technical limitations exist with the commonly practiced techniques that hinder the identification of essential interactors, which may only be present in a few copies per cell. As a result, data generated using these techniques can sometimes be biased, limiting the ability to compare these protein interactomes [20] precisely. For instance, the Y2H assay is a potent genetic way to map PPIs at both a micro-level and high-throughput manner [21, 22]. PPIs are identified in the Y2H system by stimulating reporter genes in response to a restored transcription factor [23]. Since only a few distinct yeast colonies are isolated and examined in a single experiment, the results from Y2H tests are insufficient. While fewer colonies must be evaluated, hundreds of fundamental interactions may be eliminated by a stringent requirement for interaction with binding partners for greater degrees of transcriptional activation [24]. A matrix-formatted approach can be employed to address this problem and achieve comprehensive proteome coverage. In this approach, an array consisting of an individual prey is digitally analyzed. Nevertheless, this strategy is neither readily available nor cost-effective for an individual researcher [20, 25, 26]. Alternatively, we can perform other assays using arrayed prey libraries in batches, and interacting clones can be assessed through parallel high-throughput sequencing [27]. The Y2H-based DEEPN approach has been previously developed for large-scale PPI studies in a single plate/batch format [24]. Moreover, DEEPN can eliminate most of the restrictions, including the need for infrastructure, cost-effectivity, bioinformatics expertise, instrumentation, and optimization. The details of DEEPN can be found in [24]; here layout of this previously published system in a step-by-step protocol is presented. Y2H-cDNA prey plasmids are chosen to associate with a particular bait plasmid, and DEEPN tracks their abundance in a library population. In addition, Y2H interactions can predict the relative affinities of a specific prey recombinant for one bait protein compared to another. To achieve this, it is possible to utilize various bait plasmids to target distinct prey populations within a Y2H-cDNA library with similar starting populations [24, 28]. The DEEPN approach has the advantage of identifying interactions that are strong for one bait but weak for

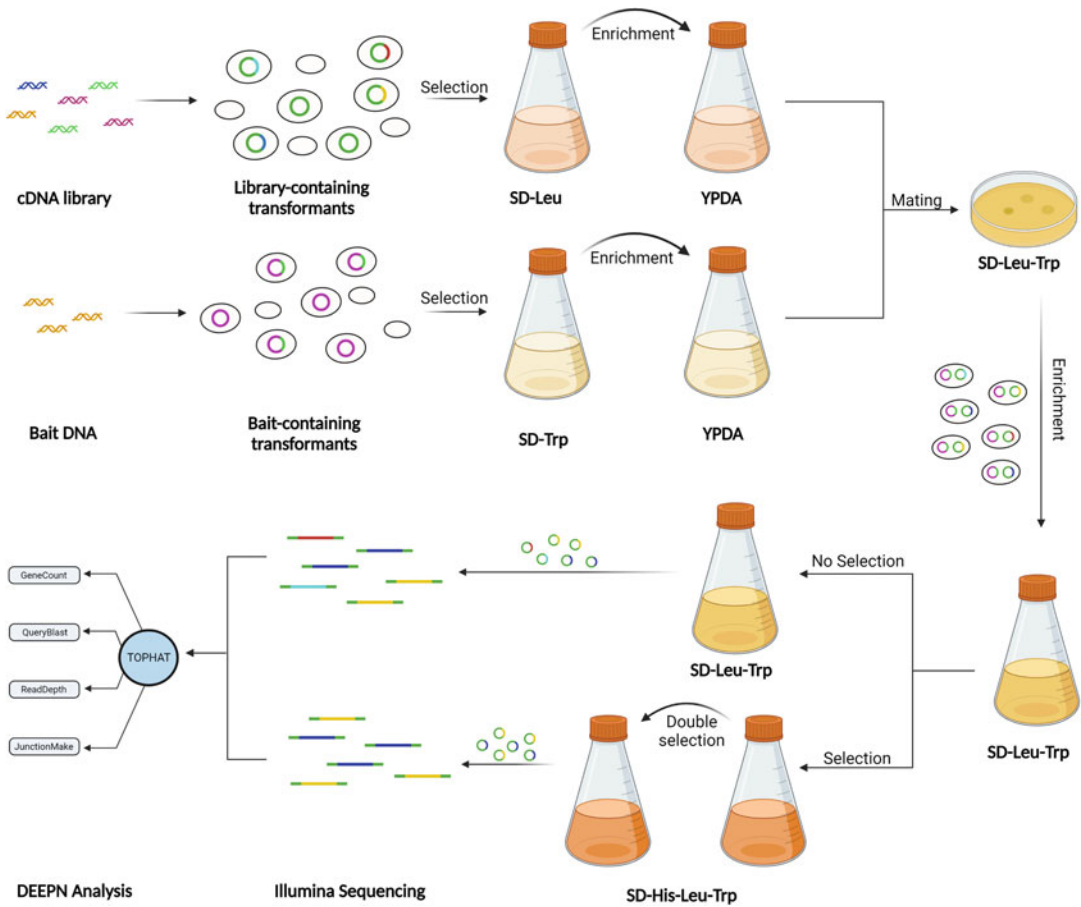


Fig. 1 Experimental schematic of DEEPN. cDNA prey library and Bait DNA are integrated into antibiotic-resistance-containing vectors, followed by transformation into respective strains. Both strains are screened with the corresponding selection medium and further enriched by a non-selection medium. After mating between prey-containing transformants and bait-containing transformants on a solid double selection medium, the diploids possessing both vectors were amplified with a similar but liquid double selection medium. Furthermore, the enriched diploids are split into two groups. In the non-selection group, the diploids are grown on the same selection medium for the detection of prey cDNA abundance. In the selection group, two more rounds of extra selection are applied to sift out strong prey-bait interactions. Both groups are sequenced and subsequently analyzed by assigned modules within DEEPN software

another (Fig. 1). Moreover, by selecting a limited read area or competition space, the population growth can be limitless since DEEPN analyzes a fixed number of reads. In addition, the *GeneCount* module determines the aligned reads for every candidate gene, indicating the degree of enrichment for that particular gene [28]. In earlier research, it was discovered that DEEPN could differentiate between interactions specific to the active, GTP-bound conformation of Rab5 and those associated with its inactive, GDP-bound conformation [24]. This protocol describes customized reagents and protocols through which DEEPN analysis can be utilized efficiently and cost-effectively.

2 Materials

2.1 *Y2H Library Construction*

Sterile inoculation loop.
 100× 15 mm Petri dish.
 YEPD Broth.
 Parafilm.
 15 mL centrifuge tubes.
 Glycerol.
 Chloroform.
 Phenol.
 Isoamyl alcohol.
 Beta-mercaptoethanol.
 Ammonium acetate.
 250 mL conical flask.
 1 L conical flask.
 96-well plate.
 Glass beads.
 DMSO.
 1.5 mL Eppendorf tubes.
 MATA PJY69 yeast.
 MAT α Y187 yeast.
 pGADT7 AD vector.
 pGBKT7 DNA-BD vector.
 EcoR1 enzyme.
 BamH1 enzyme.
 Tryptophan antibiotic.
 Yeast transformation kit.
 Synthetic-defined (SD) media.
 Yeast-rich media (YPD).
 Yeast-buffered-rich media (bYPDA).
 TWIRL sample buffer.
 Zymolyase 100 T (ThermoFisher) 10 mg/mL in Buffer (50 mM K₂PO₄ pH 7.5/50% glycerol).
 RNase A, DNase protease-free stock.
 Synthetic gene fragment.
 Gibson Assembly Master Mix (Biolabs).
 High-Fidelity DNA Polymerase 2× PCR Master Mix (Ampliqon).

Primers: to amplify inserts of mouse cDNA library in TrpI vector.
 PCR purification Kit (Monarch).

E. coli competent cells.

Normalized Mate and Plate Universal Mouse cDNA Library
 (Takarabio).

DNA Single Index Kit (Takarabio, Cat. R400697).

2.2 Antibodies

Monoclonal anti-HA antibodies (Thermo Fisher Scientific, Cat. 26183).

Polyclonal anti-myc antibodies (Thermo Fisher Scientific, Cat. R951–25).

Monoclonal 6x His-Tag antibodies (Thermo Fisher Scientific, Cat. R960–25).

Monoclonal V5-Tag antibodies (Thermo Fisher Scientific, Cat. R960–25).

Monoclonal Ubiquitin antibodies (Thermo Fisher Scientific, Cat. MA5–28514).

2.3 Special Equipment

Thermal Incubator.

Refrigerator.

Spectrophotometer.

–80 °C Freezer.

PCR machine (thermocycler).

Centrifuge.

Vortex machine.

Shaking thermal incubator.

Water bath.

Gel Documentation System.

Illumina HiSeq2500.

2.4 Composition of All Buffers, Media, and Solutions

Prepare all buffers, media, and solutions using autoclaved deionized H₂O and keep the reagents at room temperature.

2.4.1 Synthetic Defined (SD) Media for Yeast

Yeast Nitrogen Base (Ammonium sulfate) without amino acids.

Dextrose (final conc. 2%).

Adenine (200 mg/L).

Arginine (20 mg/L).

Aspartic acid (100 mg/L).

Glutamate monosodium (100 mg/L).

Histidine (200 mg/L).

Leucine (60 mg/L).
 Phenylalanine (50 mg/L).
 Serine (375 mg/L).
 Threonine (200 mg/L).
 Tryptophan (200 mg/L).
 Tyrosine (30 mg/L).
 Valine (150 mg/L).
 Uracil (200 mg/L).
 Omit Leucine.
 Tryptophan/Histidine.

1.5% Bacto Agar for solid plates.

Add all the components in a conical flask and bring up the volume to 1 L by adding deionized H₂O and sterilize by autoclaving (*see note 1*). Add bacto agar and pour into sterile Petri plates ($\approx 25\text{--}30$ mL) of autoclaved media for solid plates.

2.4.2 Yeast-Rich Media (YPD)

Peptone (20 g/L).
 Yeast extract (10 g/L).
 Sugar (dextrose/glucose) (20 g/L).
 Agar (18 g/L).

Add and mix all the components in a conical flask and bring up the volume to 1 L by adding deionized H₂O and sterilize by autoclaving. Add agar and pour into sterile Petri plates ($\approx 25\text{--}30$ mL) of autoclaved media for solid plates.

2.4.3 Buffered Yeast-Rich Media (bYPD)

Peptone (20 g/L).
 Yeast extract (10 g/L).
 Glucose (20 g/L).
 Adenine (200 mg/L).
 HCl.

Add and mix all the components (except HCl) to make the required media volume. Then add HCl to adjust the pH of the media to 3.7 and filter-sterilize it.

2.4.4 TWIRL Sample Buffer

Urea (8 M).
 SDS (4%).
 Glycerol (10%).
 Tris-HCl (50 mM).
 Bromophenol blue (0.02%).

Add and mix all the components in autoclaved deionized H₂O. Then add the HCL to adjust the pH to 6.8.

3 Methods

3.1 Development of Plasmids Containing Gal4-DNA-Binding Domain

The synthetic DNA fragment was cloned into the pGBKT7 plasmid harboring kanamycin and tryptophan-resistant genes. The vector carries a TRP1 nutritional marker for selection in yeast.

1. The synthetic DNA fragment was PCR amplified using the Gibson Assembly Master Mix following manufacturer's protocol [29].
2. Visualize the desired PCR amplified product on 1% agarose gel.
3. Purify the PCR product using the Monarch PCR purification kit.
4. Digest the Kanamycin-resistant plasmid pGBKT7 using BamHI and EcoRI restriction enzymes.
5. Mix the purified PCR synthetic DNA fragment and plasmid in Gibson Assembly Reaction and incubate in a thermocycler at 50 °C for 1 h.
6. The heat shock method was used for DNA Transformation into *E. coli* competent cells, discussed below:
7. Take out the vial of *E. coli* competent cells stored at −80 °C and thaw them on ice for 10–15 min.
8. Mix 10 ng of DNA into competent cells by pipetting while incubating it on ice for 20–30 min.
9. Put the competent cells into a pre-heated water bath/heat block at 42 °C for 45–60 s (*see* **Notes 1–6**).
10. Immediately put it back on ice for 2–5 min.
11. Pour 1 mL of LB growth media to the vial and incubate on a shaker at 37 °C for 1 h.
12. After incubation, spread the transformed cells onto an LB agar media plate having kanamycin or an appropriate antibiotic and incubate at 37 °C in a shaker overnight.
13. The following day, pick the single bacterial colony and inoculate it into selection (kanamycin-LB) media and again incubate it in a shaker at 37 °C overnight.
14. Measure the OD₆₀₀ of the culture using a spectrophotometer to ensure it falls within the range of 0.8–1.
15. Isolate the plasmid using the alkaline lysis method [30].
16. Set up the colony PCR to verify the transformed colonies. Prepare 3–5 reactions per sample by adding 50 pmoles of forward and reverse primers, 5 µg of template DNA, and 12.5 µL of High-Fidelity DNA Polymerase 2x Master Mix, and make up the final volume to 25 µL using nuclease-free water.

3.2 Expression of Fusion Proteins Containing Gal4 DNA-Binding Domain

The TRP1-carrying bait plasmids were transformed into the yeast MATA strain PJ69-4A, which is ideal for observing the yeast two-hybrid (Y2H) interactions. After transformation, the resultant strain is mated with the Y187 strain that contains the Y2H library.

1. Transform yeast cells PJ69-4A with the bait plasmid to confirm the presence of the Gal4-DNA DBD-fusion protein following the previously published protocol [21].
2. Grow transformed PJ69-4A cells overnight on 1 mL SD media lacking tryptophan (SD-Trp).
3. Next day, dilute the grown cells with two volumes of YPD media and grow for 1 h at 30 °C.
4. Centrifuge the cells and resuspend the pellet into 1 mL of 0.2 N NaOH.
5. Incubate at 25 °C for 5 min.
6. Centrifuge and re-pellet the cells, and discard the NaOH.
7. Add 100 µL of TWIRL buffer for resuspension.
8. Incubate the sample for 5 min at 70 °C for lysis.
9. Analyze/quantify the lysate using SDS-PAGE and immunoblotting [31] using anti-myc antibodies.

3.3 Self-Activation Test

3.3.1 Test Selection Conditions for Yeast Two-Hybrid (Y2H) Interaction

The Gal4-DBD-fusion proteins must be tested for circumstances that favor potential Y2H interactions. For this purpose, the bait and library prey plasmids are placed inside the same diploid background after the mating.

1. Y187 strain was transformed with empty pGBKT7 vector-only prey plasmid expressing the Gal4-transcriptional activation domain using the above protocol. For instance, the pGADT7 vector with LEU2 gene for selection in yeast and ampicillin resistance for sample in bacteria.
2. Mate the Trp + PJ69-4A carrying bait plasmid with Leu + Y187 carrying prey through patching them on the YPD plate.
3. Incubate the plates overnight at 30 °C.
4. Next day, streak the yeast cells using a sterile loop on the SD media plate lacking both Leu and Trp.
5. Isolate the single PJ69-4A/Y187 Mata/diploid colonies from the SD-Leu plate.
6. Make testing plates with different combinations: (i) SD-Leu-Trp, SD-Trp-Leu-His and (ii) SD-Leu-Trp-His with 0.1 to 10 mM 3AT (3-amino-triazole).
7. Patch the diploids on the plates and let them grow overnight.
8. Pellet and resuspend the grown cells in sterile water.

9. Measure the OD and maintain it at $OD_{600} = 0.5$ by further resuspension (if required).
10. Perform the serial dilution at 1:10.
11. Spot 4 μL from each dilution on pre-prepared test plates (*see note 2*).
12. Let the cells grow by incubating the plates at 30 °C for 2–3 days.

3.3.2 Mating and Selection

It can be difficult to mate with the Y187 strain. Therefore, the following optimized protocol must be followed to preserve the library's complexity.

• Day 1

1. Inoculate the (*Trp*) pGBKT7 bait plasmid carrying PJ69-4A yeast strain into 23 mL of freshly prepared SD-Trp media.
2. Inoculate Y187 yeast cells with LEU2-carrying “prey” library into 125 mL of freshly prepared SD-Leu media.
3. For overnight growth, put the cultures in a 30 °C shaker incubator at 200 rpm.

• Day 2

1. Measure the OD_{600} , which should be between 1.0 and 1.5.
2. Centrifuge and pellet the cells of the Y187 strain containing the library plasmids and PJ69-4A cells in separate 50 mL sterile conical tubes.
3. Resuspend the cell pellet in 10 mL of sterile water.
4. Transfer into new 50 mL sterile culture tubes and re-pellet by centrifugation.
5. Resuspend the PJ69-4A and Y187 cells into the 4 mL of bYPD media separately.
6. Adjust the pH to 3.7 by adding 1 N HCl.
7. For mating, take a sterile 50 mL conical tube and add 1 mL of each, i.e., PJ69-4A, Y187, and bYPD, into it.
8. For incubation, put the reaction mixture in a shaker incubator at 30 °C at 90–130 rpm for 90 min (*see Note 3*).
9. Centrifuge the cells and resuspend the pellet into 2 mL of bYPD.
10. Spread the entire 2 mL of bYPD resuspension on a 150 mm YPD-agar plate and overnight incubate at 30 °C.

• Day 3

1. Harvest the cells using a cell scraper and rinse with 10–12 mL of SD-Leu-Trp media.
2. Inoculate the harvested cells from the YPD plates into 12 mL SD-Leu-Trp media in a 50 mL sterile conical tube.

3. Pellet the cells by centrifugation and resuspend the pellet into 40 mL of SD-Leu-Trp media.
4. To measure the diploid cells in the reaction mixture, dilute 4 μ L of sample into 200 μ L of SD-Trp-Leu media.
5. Gently mix by pipetting and plate it onto an SD-Leu-Trp selection plate and incubate at 30 °C for two days.
6. Inoculate the remaining 40 mL of resuspension culture into 500 mL of SD-Leu-Trp media.
7. Incubate the flasks for 36 h at 30 °C (or more) in a shaker/incubator at 200 rpm until the OD₆₀₀ reaches 2.0.

- **Day 4**

1. Observe the growth in each flask by measuring the OD₆₀₀.

- **Day 5**

At this stage, we split the yeast culture into two for selection and non-selection. The titer plates ought to be ready for analysis by this time, enabling verification of enough mating efficiency to sustain library complexity (*see* **Note 4**).

1. Take the 20 mL of grown culture and inoculate it into 750 mL of SD-Leu-Trp (non-selection) media.
2. Take another 20 mL of grown culture and inoculate it into 750 mL of SD-Trp-Leu-His (selection) media with minimal 3AT concentration (*see* **Note 5**).
3. Gently mix the new 770 mL cultures and measure the initial OD₆₀₀.
4. Let the cultures grow separately in the shaker at 30 °C at 200 rpm until it reaches the OD₆₀₀ of 2.0.

- **Day 6**

1. Confirm that the OD₆₀₀ of non-selection media is at 2.0.
2. Take 10 mL of culture and pellet it by centrifugation. Cells can be stored at –20 °C or proceed for sequencing at this stage.

- **Day 7**

1. Measure the OD₆₀₀ of the SD-Leu-Trp-His (selection) media at 2.0.
2. Take 2 mL out and inoculate into 75 mL of freshly made SD-Leu-Trp-His media with minimum 3AT to avoid background growth.
3. Let the cells grow until they reach OD₆₀₀ of 2.0 by placing them in a shaker incubator at 30 °C at 200 rpm for 30–60 h.

- **Days 8 and 9**

1. Check the cell growth by measuring the saturation of selection (SD-Leu-Trp-His) cultures at OD₆₀₀ of 2.0.
2. Take 10 mL of culture and pellet it by centrifugation. Cells can be stored at −20 °C or proceed for sequencing at this stage.

3.4 Preparation of DEEPN Samples for Illumina Sequencing

3.4.1 DNA Extraction/Purification

1. Gently resuspend pellets into 500 µL of 50 mM Tris and 20 mM EDTA.
2. Transfer it to a 1.5 mL sterile Eppendorf tube.
3. Add and mix 10 µL of Zymolase stock and 3 µL of Beta-Mercaptoethanol.
4. Incubate the sample at 37 °C for 24–36 h.
5. Add a mixture of phenol/chloroform/isoamyl alcohol (25:24:1) and carry out ethanol precipitation.
6. Add 100 µL of 50 mM Tris/20 mM EDTA in the precipitated pellet and resuspend it.
7. Add 1 µL of RNaseA from stock and incubate for 1 h at 37 °C.
8. Precipitate it again using ethanol and resuspend it into 100 µL of 5 mM Tris/2 mM EDTA.
9. Quantify the DNA using a spectrophotometer at 280 nm absorbance (*see Note 6*).

3.4.2 PCR Purification of cDNA Fragments

1. Set up two individual PCR reactions for selection and non-selection samples.
2. Set up the PCR for each reaction mixture. Add 1 µL of forward and reverse primers, 5 µg of template DNA, and 25 µL of PCR master mix, and make up the final volume to 50 µL using nuclease-free water.
3. Run the agarose gel electrophoresis on the PCR product (as discussed above).
4. For PCR product purification, the Monarch PCR purification kit was used.

3.4.3 Illumina Sequencing

1. For Illumina sequencing, shear the 400–550 ng PCR-purified product into 300 bp average length fragments using Covaris E220 Ultrasonicator.
2. Indexed Sequencing libraries were generated using the DNA Single Index Kit for the Illumina sequencing.
3. Pool up the Indexed libraries and run the samples on Illumina HiSeq2500.

3.5 DEEPN

DEEPN is a novel approach to analyzing the protein–protein interaction in Y2H (*see* **Note 7**). DEEPN is created to handle and interpret sequencing data from the Illumina platform, which generates reads between 110 and 140 bp. The distinct sides of a paired-end sequence are treated as two unique sequences by DEEPN, so both single and paired-end sequences are acceptable. Process the .fastq sequence files using Tophat2, which produces the mapped and unmapped sequence files in SAM format. DEEPN requires the mapped and unmapped sequence files to analyze and process the data. DEEPN analyzes the dataset using the following modules:

.fastq! Tophat2/Bowtie! Gene Count! Junction Make! Blast Query! Read Depth.

Gene Count is used to evaluate each gene's number of reads, wherein *Stat Maker* ranks these evaluated reads. While screening of unmapped reads from the junction between Gal4 AD and cDNA fragment is carried through *Junction Make*. To identify the insert fragment, it also runs the *Blastn*. Query Blast is used to determine the frequency and position of each junction within a gene. At last, *Read Depth* generates a plot for the mapped data of a cDNA fragment.

4 Notes

1. Filter-sterilized amino acids can be added after autoclaving when the media cools to 50 °C.
2. For the best interaction, observe the growth in the presence of His regardless of 3AT.
3. This is necessary to prevent the cells from sedimentation.
4. A minimum of one million total diploids are advised for this workflow.
5. Normally, 3AT prevents yeast growth and its concentration should be between 0.1 and 1 mM.
6. 5 µg of DNA is required for PCR amplification.
7. Several other screening experiments have been devised concurrently with the technical development of the Y2H system. Y2H libraries can be created using genomic DNA, cDNA, normalized cDNA, full-length cDNA, or open reading frame (ORF) libraries, depending on the research's goals and available materials [32]. The development of mating type a and type two-hybrid strains with the bait and prey plasmids is an alternative strategy to the original Y2H [33]. In modern Y2H testing, the mating type a is still extremely popular. Moreover, complementary assays such as YFP fluorescent-based protein complementation assay can be used as a validation technique.

The fluorescence-activated cell sorting can restore functional YFP if the two proteins interact. The nucleic acid programmable protein array (NAPPA) is another validation assay. For instance, two proteins were expressed in a linked transcription/translation reticulocyte lysate and fused to a glutathione S-transferase (GST) tag and a HA epitope tag, respectively [34]. An anti-GST antibody fixed at the bottom of a 96-well microtiter plate captures the GST-tagged protein [35, 36].

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